Energy Informatics for Facilitating Large-Scale Research Applications

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THE PROBLEM

Energy research needs weather records, grid geometry and operational data that sit in different places, in different formats. Most projects join them with one-off scripts that nobody else can rerun. This thesis proposes a reusable 9-layer data architecture and tests it on one case study.

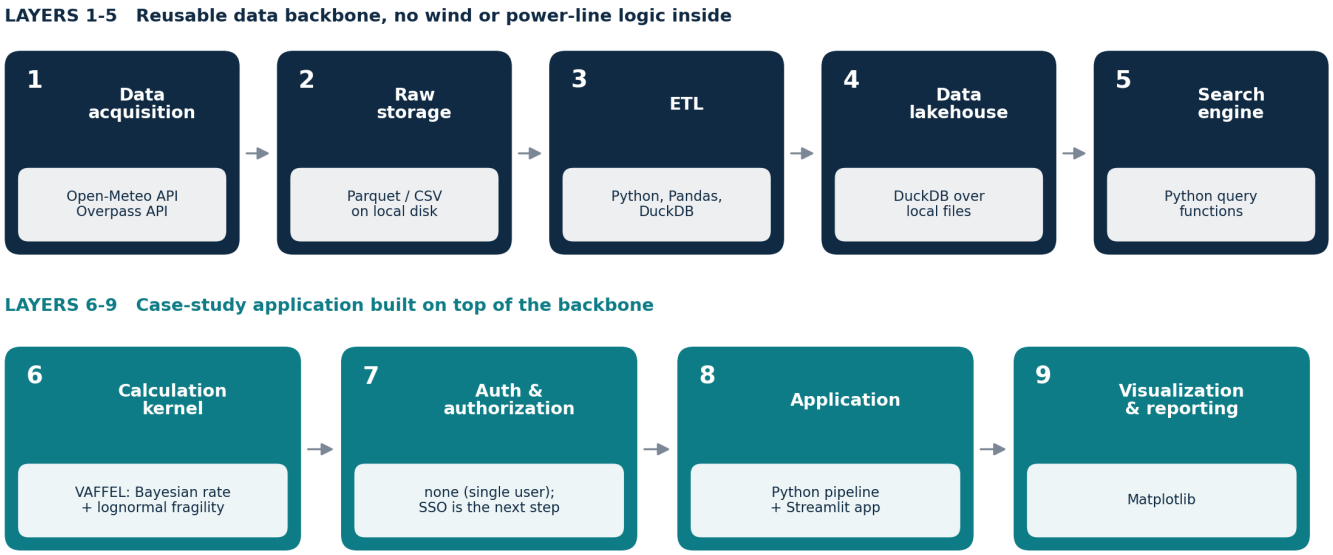


Figure: the nine layers and the tool used for each in the prototype. Alternatives and a production path are compared in the thesis.

6.7x

faster single-field queries with a structured table than with raw Parquet

Layer 4 benchmark on the ~1 GB wind dataset (DuckDB, one laptop, mean of 3 runs): 3.6 ms against 24.1 ms. The price is a one-time load of 8.7 s. For a joined query the two options are equal (61 ms against 60 ms), so the design choice depends on the query pattern.

THE CASE STUDY: WIND-DEPENDENT FAILURE OF TRANSMISSION LINES IN SICILY

Eight 220 kV and 380 kV overhead lines, hourly wind speed at 100 m from 2013 to 2023 (ERA5-Land through Open-Meteo), line and tower geometry from OpenStreetMap. The method is VAFFEL: a Bayesian update of each line's failure rate, a lognormal fragility curve fitted to it, and an hourly probability of failure for every line. Every step of the code maps to one of the nine layers, so the backbone (layers 1-5) contains nothing specific to wind.

HONEST LIMITS

The failure records are synthetic and statistically plausible, so the probabilities show that the method works from raw data to result. They are not a real risk estimate. The pipeline runs on one machine. With real operator outage records, only layer 6 would need to change.

Stack: Python, Pandas, DuckDB, SciPy, SALib, Matplotlib, Streamlit. Code: see the repository README.